

MAHARAJA INSTITUTE OF TECHNOLOGY MYSORE

CENTRE OF EXCELLENCE (CoE)

3-Day Internship Work Plan Template

Centre of Excellence: EARTH SYSTEMS AND GEOSPATIAL SCIENCE & TECHNOLOGY (ESGST)

Department: CIVIL ENGINEERING

Faculty Coordinators: Prof. MANOJ B.R. & Prof. MANJUNTHA R.N.

Batch/Branch: CIVIL **Duration:** 3 Days

3-Day Internship Program on Digital Elevation Model (DEM) Processing and Terrain Analysis Using QGIS

DAY 1 – DEM Data Acquisition and Preparation

Time	Session Title	Topics / Concepts Covered	Practical / Activity	Resources / Equipment Required	Faculty
09:00–11:15	Introduction to DEM and Terrain Data	Basics of DEM, Types of DEM (SRTM, ASTER, Copernicus), Applications of DEM in Civil Engineering, Hydrology & Watershed Management	Introduction to QGIS Interface	Laptop, QGIS, Internet, Projector	MBR & MRN
11:30–01:15	Downloading DEM and Administrative Boundary Data	Sources of DEM Data (USGS Earth Explorer, Open Topography), Downloading District Boundary Data from Survey/Open Government Sources	Download DEM Tiles and District Boundary Shapefile for Selected District	Laptop, Internet, QGIS	MBR & MRN
02:00–04:00	Data Organization and Visualization	Coordinate Systems, Layer Management, Attribute Information, DEM Visualization	Load DEM and District Boundary into QGIS and Verify Data	Laptop, QGIS, Sample Dataset	MBR & MRN

DAY 2 – DEM Pre-processing

Time	Session Title	Topics / Concepts Covered	Practical / Activity	Resources / Equipment Required	Faculty
09:00–11:15	Coordinate Reference Systems (CRS) and Reprojection	Geographic vs Projected Coordinate Systems, CRS Selection, Reprojection Concepts	Reproject DEM and District Boundary to Common CRS	Laptop, QGIS	MBR & MRN
11:30–01:15	DEM Mosaicking	Need for Merging Multiple DEM Tiles, Raster Processing Tools	Merge Multiple DEM Tiles into a Single Raster	Laptop, QGIS	MBR & MRN
02:00–04:00	Clipping DEM to Study Area	Raster Extraction Techniques, Mask Layers, Study Area Preparation	Clip DEM Using District Boundary and Prepare Final DEM	Laptop, QGIS	MBR & MRN

DAY 3 – Terrain Analysis Using DEM

Time	Session Title	Topics / Concepts Covered	Practical / Activity	Resources / Equipment Required	Faculty
09:00–11:15	Slope and Aspect Analysis	Terrain Derivatives, Slope, Aspect and Their Applications	Generate Slope and Aspect Maps	Laptop, QGIS	MBR & MRN
11:30–01:15	Hillshade and Terrain Visualization	Hillshade Generation, Terrain Visualization Techniques, Cartographic Representation	Generate Hillshade Map and Terrain Visualization	Laptop, QGIS	MBR & MRN
02:00–04:00	DEM-Based Terrain Products and Map Layout	Elevation Profiles, Contours, Map Composition, Exporting Maps	Prepare Slope, Hillshade and Final Terrain Maps with Layout and Export	Laptop, QGIS	MBR & MRN

Expected Outputs

- ☐ Attendance Sheet
 ☐ Mini Project / Prototype
 ☐ Internship Report
☐ Student Feedback
 ☐ Completion Certificate

Faculty Coordinator	HoD	Institute CoE Coordinator	Principal
<i>Signature</i>	<i>Signature</i>	<i>Signature</i>	<i>Signature</i>